

Experiment No: 1

Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required: Visual Studio Code (or any text editor), Web Browser (Google Chrome, Microsoft Edge, Firefox, etc.)

Theory:

CSS (Cascading Style Sheets) is a style sheet language used to control the appearance and layout of web pages. It separates the presentation of a webpage from its structure, making websites more attractive, consistent, and easier to maintain.

There are three ways to apply CSS to an HTML document:

1 . Inline CSS

- CSS is written directly inside an HTML element using the style attribute.
- Suitable for applying styles to a single element.
- Example: `<p style="color: blue;">Hello</p>`

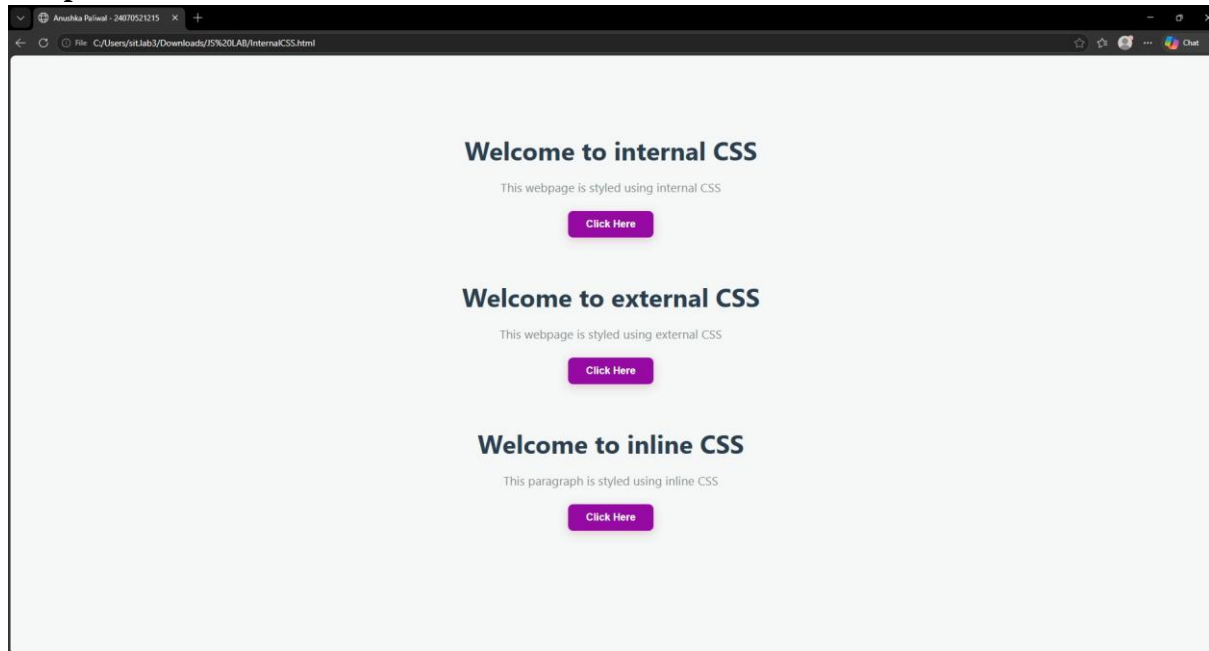
2. Internal CSS

- CSS is written inside the `<style>` tag within the `<head>` section of an HTML document.
- Used when the styles are specific to a single webpage.
- Helps keep styling separate from HTML content within the same file.

3. External CSS

- CSS is written in a separate .css file and linked to the HTML document using the `<link>` tag.
- Makes styles reusable across multiple webpages.
- Improves code organization, maintainability, and consistency throughout a website.

Output:



Result/Conclusion:

The experiment was successfully completed. The three methods of applying CSS—Inline, Internal, and External CSS—were implemented and understood. It was observed that External CSS is the most efficient and maintainable approach for styling multiple webpages, while Inline and Internal CSS are useful for small-scale or page-specific styling.

CASE STUDY

Experiment Title:

Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required:

Text Editor (Visual Studio Code, Notepad++, etc.)

Web Browser (Google Chrome, Mozilla Firefox, Microsoft Edge, etc.)

Experiment Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initialscale=1.0">

    <title>SIT Nagpur - IT Department</title>

    <!-- Internal CSS for styling matching the template -->

    <style>
body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

    background-color: #f3f4f6;

display: flex;        justify-content: center;
align-items: center;   height: 100vh;
margin: 0;

    }
    .card
    {
        background: white;        padding:
40px;        border-radius: 12px;

        box-shadow: 0 4px 15px rgba(0, 0, 0, 0.05);        text-align:
center;        max-width: 500px;        width: 100%;

        border-top: 8px solid #064e3b;

    }
h1 {
    color: #1f2937;        font-size:
2.5rem;        margin-bottom: 5px;

```

```
    }  
h2 {  
    color: #065f46;  
    font-size: 1.3rem;    font-weight: 500;  
margin-top: 0;    margin-bottom: 25px;  
}
```

```
p {  
    color: #6b7280;    font-size: 1rem;  
line-height: 1.6;    margin-bottom: 30px;  
}    .btn  
{
```

```
    background-color: #0ea5e9;  
color: white;    border: none;  
padding: 12px 28px;    font-size: 1rem;  
font-weight: 600;    border-radius: 6px;  
cursor: pointer;  
    transition: background-color 0.2s ease;  
}
```

```
.btn:hover {  
    background-color: #0284c7;  
}
```

```
</style>
```

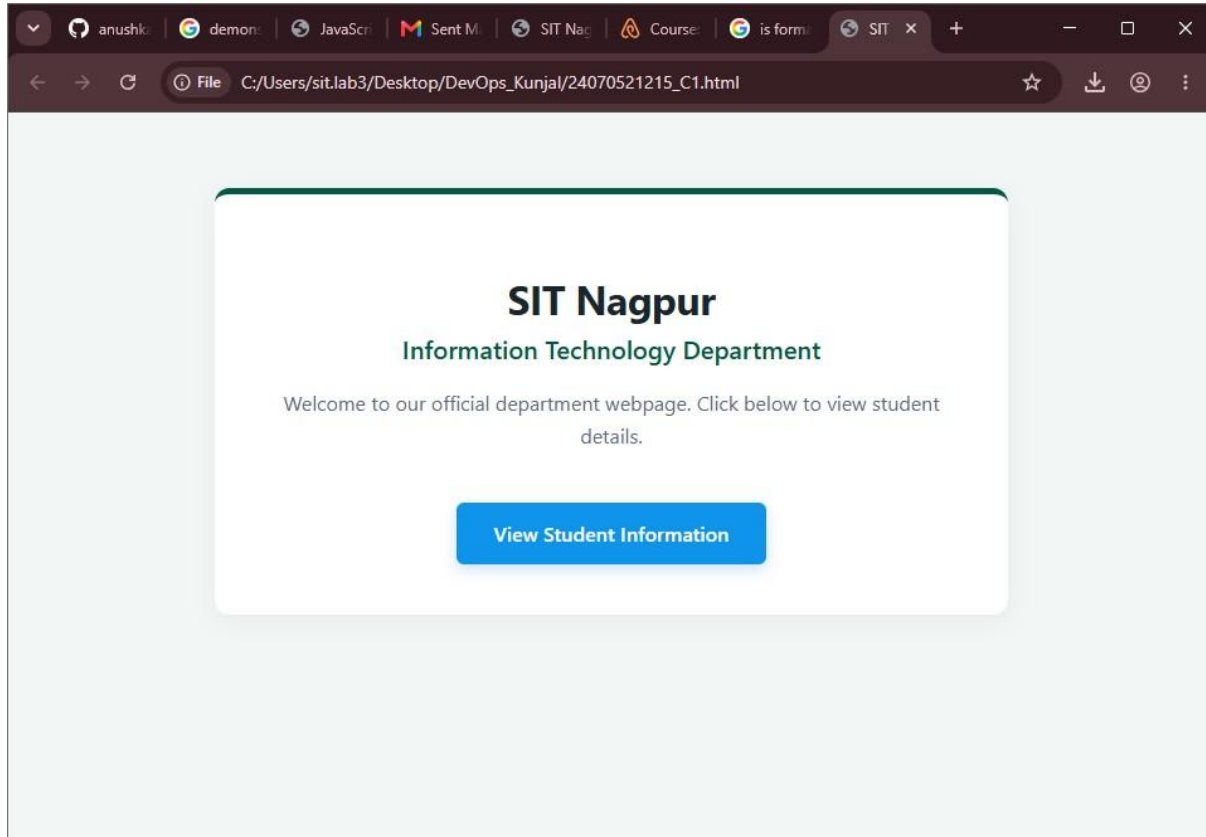
```
<!-- Internal JavaScript -->
```

```
<script>
```

```
    // Console methods demo inside internal script    console.log("Internal JS loaded.  
Webpage initialized.");    console.error("This is a simulated system check tracking errors  
(None found).");
```

```
// Welcome message on window load      window.onload =  
function() {  
    console.log("Webpage fully loaded and rendered.");  
};  
</script>  
</head>  
<body>  
  
    <div class="card">  
        <h1>SIT Nagpur</h1>  
        <h2>Information Technology Department</h2>  
        <p>Welcome to our official department webpage. Click below to view student  
details.</p>  
  
        <!-- Inline JavaScript to trigger external function and a console tracking log -->  
        <button class="btn" onclick="console.log('Button clicked.');" displayStudentInfo();">  
            View Student Information  
        </button>  
    </div>  
  
    <!-- Linking External JavaScript -->  
    <script src="script.js"></script>  
</body>  
</html>
```

Output



Result/Conclusion:

The structural setup demonstrates the integration of multiple JavaScript deployment styles successfully. Inline execution handled direct event tracking, internal scripting captured loading workflows with console diagnostics (`log`, `warn`, `error`), and external logic kept data processing code clean and modular. The final webpage accurately mirrors the standard department interface design while executing basic functional feedback objects.